

ASSIGNMENT-03

AIM :- Design an interactive front end application using React by implementing templating using components, states & Props, class, Events. It must be responsive to scale across different platforms.

OBJECTIVE :- To develop a responsive, interactive front-end application using React.js that effectively demonstrates component-based architecture, state management, & event handling with scalable UI components, dynamic data via states & props, & seamless user interactions across devices.

THEORY :-

> Explain the role of State & Props in React. How do they differ, & what is the primary purpose of each in managing data flow within a component-based application.

State :- Represents mutable data owned & managed by a component. It allows components to create dynamic & interactive UI's by updating themselves when the state changes.

Props :- Short for 'properties', props are read-only inputs passed from a parent component to a child, enabling data sharing across components.

Difference :- State is internal & changeable, while props are external & immutable. Together, they enable unidirectional data flow in react.

Q2 What is a React component? Differentiate between a class component & a functional component, & discuss the advantages of using a functional component with hooks like `useState` & `useEffect` over a class component.

Ans - Component : A reusable, independent piece of UI in React.
Class Components : Defined using ES6 classes, use `this.state` & lifecycle methods.

Functional Components :- Defined as functions, use hooks like `useState` & `useEffect` for state & lifecycle management.

Advantages of Functional + Hooks :- Cleaner syntax, less boilerplate, better performance, easier to test, & modern React development favors hooks over classes.

Q3 Describe the concept of "templating using components" in React. Why is this approach considered superior to traditional web development methods that rely on monolithic HTML files?

Ans - Breaks UI into reusable, modular components.

• Superior to monolithic HTML → improves reusability, scalability & maintainability.

Q4 How do you handle user events in React? Provide a simple code snippet to demonstrate how an event handler is defined in a component & how it can be used to update the component's state.

Ans - Handlers defined as functions; update state using `useState`.

```
import React, { useState } from "react";
function Counter() {
  const [count, setCount] = useState(0);
  return (
    <div>
      <p>{count}</p>
      <button onClick={() => setCount(count + 1)}>Inc </button>
    </div>
  );
}
```

Q.5 What is responsive design, why is it crucial for modern applications? Describe how you would implement a responsive design in a React application using CSS media queries or a CSS-in-JS library.

Ans - Ensures UI adapts to all screen sizes/devices.
Implement using CSS media queries or CSS-in-JS
Eg :- `@media (max-width: 600px) {`

{ font-size: 14px; }
 }

Ans
23/9/25